Federated Multimodal Sentiment Analysis Under Missing Modalities and Non-IID Skew: An Empirical Investigation of Modality-Aware Aggregation and Information Balancing

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Abstract

Multimodal sentiment analysis aims to understand and recognize human emotions by fusing text, speech and facial expressions. With the aid of federated learning (FL), multiple devices can collaboratively train a shared model without sending the raw audio/video data to the central server. However, there are three challenges when applying FL to real applications on edge devices. First, each client does not have access to all modalities which results in some missing modalities. Second, when one modality is more informative than others, the model may focus on the informative modality and fail to learn from the less informative modalities. Third, the amount of data and distribution of labels may differ greatly between clients, resulting in high non-IID data.

This paper presents an empirical study on FIB-Fusion which is a federated framework consisting of three fundamental elements decoupled linear-logit fusion, modality aware block-wise server aggregation, and Information Balanced Multimodal Learning (IBML) including Balance Information Objective (BIO) and Task Complexity Modulation (TCM).

The proposed framework is validated with ten simulated edge clients, CMU-MOSI and CMU-MOSEI benchmark datasets. The experiments are performed with three different seed values. Results are provided for seven scenarios (TAV, TA, TV, AV, T, A, V). The modality aware aggregation method not only avoids effects of missing channels on the updates of the available channels but also reduces communication overhead by 18.35

FedProx+IBML achieves the best performance in CMU-MOSEI. It obtains a test macro-F1 of 0.7439 ± 0.0198 in full modality setting, and an average 0.6645 ± 0.0156 in partial modality setting. It outperforms centralized baselines. In CMU-MOSI, the best federated model is obtained by using cross-entropy combined with BIO, under modality-aware aggregation strategy, achieving 0.7555 ± 0.0160, which is very close to the cross-entropy centralized baseline. However, IBML-based models do not achieve such good performance in MOSI. This could be due to the challenging nature of balancing information with clients of small minibatch sizes and highly imbalanced label distribution.

Keywords: Federated Learning, Multimodal Sentiment Analysis, Missing Modalities, Information-Balanced Multimodal Learning, Non-IID Data Distribution, CMU-MOSEI, CMU-MOSI.

1 Introduction

Sentiment analysis on spoken opinion videos has many applications within affective computing, such as human computer interaction, gaining insight into customer experience and digital healthcare monitoring [1, 2]. During conversation people express their emotions in various ways. The tone and features in the voice such as pitch, jitter and energy can help interpret sentences. People also show their emotions via facial expressions such as smiling or frowning. Generally combining the modalities will produce improved results. However, this can only be done effectively if all modalities are available and aligned.

In practice, we often store data locally and may have concerns about sharing raw video/audio data with a central server. Federated learning is an effective solution as it enables the models to be trained locally on user devices, only sharing parameters of the model itself [3, 4]. There are still some challenges that need to be overcome:

1. **Client-level missing modalities.**

   Depending on the scenario, not all users may have the same rights and/or hardware. This may mean that different modalities (cite shah2023cite park2022) are available to each client. For example one client may not have access to the camera but may have text transcripts. Another client may only have access to audio recordings. The problem here is that FedAvg assigns equal weightings to all parameters. This causes clients without the modality to contribute towards its update.

2. **Modality optimization imbalance.**

   In practice, during training, multi-modal models may rely heavily on the fast learning of text modality, leading to poorly trained acoustic and visual encoders [wang2020,qin2026]. While we are able to alleviate this issue via IBML during central training, it is unclear whether our approach will be effective when the client mini-batch sizes are small and unbalanced.

3. **Statistical heterogeneity (non-IID skew).**

   Affective data can vary greatly between different speakers and contexts [zhao2018]. Large differences in local label distributions cause local updates to drift away from the global objective.

We address these issues by introducing the FIB-Fusion framework that uses decoupled linear-logit fusion, modality-specific block-wise aggregation, and adapted BIO/TCM approaches to learn from federated mini-batches. We evaluate our framework under 7 full and degraded scenarios, namely TAV, TA, TV, AV, T, A, and V.

2 Related Work

2.1 Multimodal Sentiment Analysis

Sentiment analysis has moved from using basic lexical n-gram features to deep multimodal representations [zadeh2016, zadeh2018]. The standard benchmarks used for this task are CMU-MOSI (Multimodal Opinion-level Sentiment Intensity) and CMU-MOSEI (Multimodal Opinion Sentiment and Emotion Intensity). Advanced architectures, e.g., Multimodal Transformer (MulT) [tsai2019] and dynamic fusion graphs have been developed to better understand cross-modal correlations. Most of these approaches assume that all modalities are available during training and testing.

2.2 Federated Optimization and Non-IID Data

Federated Averaging (FedAvg) [3] trains a global objective function on clients by averaging the weights of the locally trained parameters using a weighted scheme. For clients with different data distributions, the local models will diverge. To address this issue, a proximal L2 penalty is added to the local objective function to restrict the divergence in FedProx [4].

2.3 Missing Modalities and Modality Imbalance

Missing modalities are often resolved using autoencoder reconstruction, cycle translation or modality drop out [6, 11]. Federated learning has been explored with respect to multimodal sentiment analysis by Shah et al. [5] within a low resource client setting.

Qin et al. [8] recently proposed a Deep Multimodal Information-Balanced Learning (IBML), which demonstrates that introducing auxiliary multi-task balancing losses and perturbations in features prevents a dominant modality from suppressing weaker modalities during centralized learning.

We present balancing mechanisms in a federated setting with heterogeneous clients.

3 Problem Formulation

Let us consider a federated network with $K = 10$ edge clients orchestrated by a central parameter server. Each client k having $k \in \{1, \ldots, K\}$ has its own private local dataset:

$$D_k = \left\{(x_i^T, x_i^A, x_i^V, y_i)\right\}_{i=1}^{n_k}. \tag{1}$$

Here:

- $x_i^T \in \mathbb{R}^{D_T}$ holds the text features.
- $x_i^A \in \mathbb{R}^{D_A}$ holds the audio features.
- $x_i^V \in \mathbb{R}^{D_V}$ holds the visual features.
- $y_i \in \{0, 1\}$ is the binary sentiment label.

The total number of samples across clients is:

$$N = \sum_{k=1}^{K} n_k. \tag{2}$$

Each client also has a modality availability mask:

$$M_k \subseteq \{T, A, V\}. \tag{3}$$

Client k computes the forward pass using only available modalities in M_k. Modalities not available are ignored and not encoded.

We seek to find a unified set of parameters Θ that minimize the global objective

$$\min_{\Theta} F(\Theta) = \sum_{k=1}^{K} \frac{n_k}{N} F_k(\Theta; M_k). \tag{4}$$

4 Proposed Methodology

FIB-Fusion consists of three components: local decoupled features, server-side modal-aware cross-modal aggregation and client-side info-balancing.

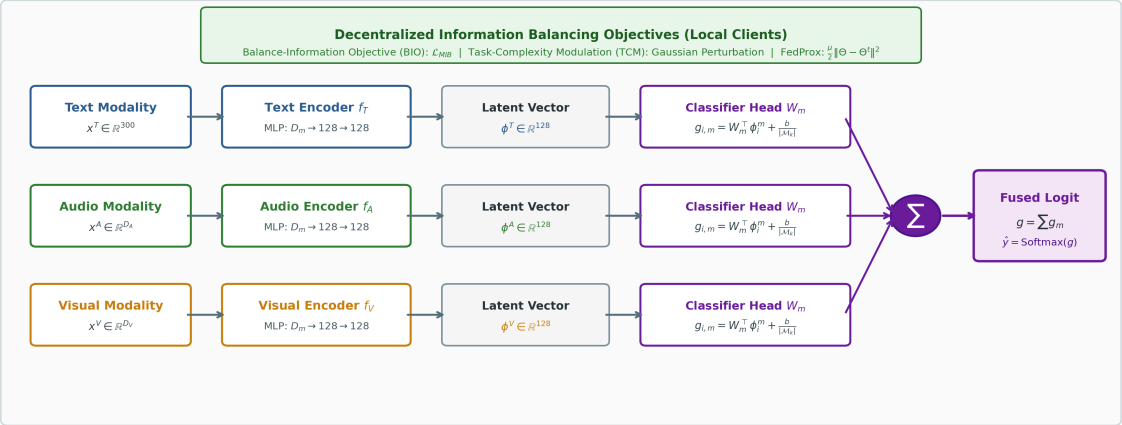


Figure 1: Overall System Architecture For Our FIB-Fusion Framework

The modality-specific encoders project the features of the activated sensors into the common latent space using f_T, f_A and f_V. These are then transformed by modality-specific heads with a shared bias. While all heads take the latent features as input, they contain their own specific weights W_m. The linear classification bias is shared equally between all heads. This is done by adding a bias term $b/|M_k|$ to each head. The final stack of logits is just the summation of the predictions coming from the activated modalities, g. It can be seen that the b is recovered exactly. This makes sense in the case of additional supervision signals which help train the model during the local training phase (e.g. Protect-Information Objective (BIO), Task-Complexity Modulation (TCM)).

4.1 Modality-Specific Encoders

For each modality $m \in \{T, A, V\}$, one dedicated encoder $f_m(x; \theta_m^{\text{enc}})$ is defined. If no channel is present, other modalities are left unmodified. Given the input data $x^m \in \mathbb{R}^{D_m}$ we map it to the latent space $z \in \mathbb{R}^{128}$ via the encoder:

$$\phi^m = f_m(x^m). \tag{5}$$

It applies these operations in order:

- A linear transformation from the modality-specific input dimension to 128 dimensions.
- ReLU activation.
- Dropout with probability $p = 0.2$.
- A second linear transformation from 128 to 128 dimensions.
- ReLU activation.

The weight matrices have these shapes:

$$W_{m,1} \in \mathbb{R}^{128 \times D_m}, \tag{6}$$

$$W_{m,2} \in \mathbb{R}^{128 \times 128}. \tag{7}$$

4.2 Decoupled Linear-Logit Fusion

We use a linear-logit decomposition (LinearIBMLNet) to facilitate attribution of contributions from each modality:

Each active modality $m \in M_k$ will have an unbiased linear head $W_m \in \mathbb{R}^{128\times 2}$ that transforms the latent representation into class logits. We assign the bias vector ($b \in \mathbb{R}^2$) equally to all the active modalities.

$$g_{i,m} = W_m^T \phi_i^m + \frac{b}{|M_k|}. \tag{8}$$

This is the sum of the split bias across all active branches:

$$g_i = \sum_{m\in M_k} g_{i,m}, \tag{9}$$

which gives:

$$g_i = \sum_{m\in M_k} W_m^T \phi_i^m + b. \tag{10}$$

This results in the exact reconstruction of b as:

$$\sum_{m\in M_k} \frac{b}{|M_k|} = b. \tag{11}$$

Inactive modalities contribute nothing to the fused logit and hence receive no gradient during back-propagation.

4.3 Local Client Optimization and Objectives

For a local minibatch B drawn from D_k, the task cross-entropy loss on the fused prediction is:

$$\mathcal{L}_{CE}(g, y) = -\frac{1}{|B|} \sum_{i\in B} \log \left[\frac{\exp(g_{i,y_i})}{\sum_{c=0}^{1} \exp(g_{i,c})} \right]. \tag{12}$$

4.3.1 Balance-Information Objective (BIO)

We calculate the total softmax confidence for each active branch, as proposed by Qin et al. [8]:

$$\rho_m = \sum_{i\in B} [\text{Softmax}(g_{i,m})]_{y_i}. \tag{13}$$

The mean confidence is:

$$\bar{\rho} = \frac{1}{|M_k|} \sum_{m\in M_k} \rho_m. \tag{14}$$

Where I_m is an indicator variable (1 if $\rho_m < \bar{\rho}$, 0 otherwise) used to identify modalities whose confidence is less than the average.

With:

$$\hat{I} = \max\left(1 - \sum_{m} I_m, 0\right), \tag{15}$$

the multi-task balance loss is:

$$\begin{aligned}\mathcal{L}_{MIB} = \hat{I}\mathcal{L}_{CE}(g, y) \\ + \sum_{m \in M_k} I_m \left[\mathcal{L}_{CE}(g, y) + \mathcal{L}_{CE}(g_m, y)\right].\end{aligned} \tag{16}$$

For a unimodal client ($|M_k| = 1$), there are no other modalities to compare to and the loss becomes simple cross entropy:

$$\mathcal{L}_{CE}(g, y). \tag{17}$$

The local loss for this joint BIO and CE formulation (Method A6) is,

$$\mathcal{L}_{local} = \mathcal{L}_{CE}(g, y) + \lambda_{BIO}\mathcal{L}_{MIB}, \tag{18}$$

where:

$$\lambda_{BIO} = 0.5. \tag{19}$$

4.3.2 Task-Complexity Modulation (TCM)

where the clients maintain a log of which samples have been correctly classified.

Dominant modalities (those with $\tilde{\rho}_m \geq \bar{\rho}$) perturb features of correctly classified samples using a Gaussian distribution which is scaled by their feature variance. The probability of doing so is,

$$p_m = \frac{\min\left(M\frac{\tilde{\rho}_m}{\sum_j \tilde{\rho}_j}, \lambda\right)}{\lambda}. \tag{20}$$

The scaling constant is:

$$\lambda = 10. \tag{21}$$

4.3.3 FedProx Regularization

A proximal penalty around the global parameters Θ^t from the current round is applied during local training to prevent clients drifting too far away when working with non-IID partitions.

$$\mathcal{L}_{prox}(\Theta) = \mathcal{L}_{local}(\Theta) + \frac{\mu}{2}\left\|\Theta - \Theta^t\right\|_2^2. \tag{22}$$

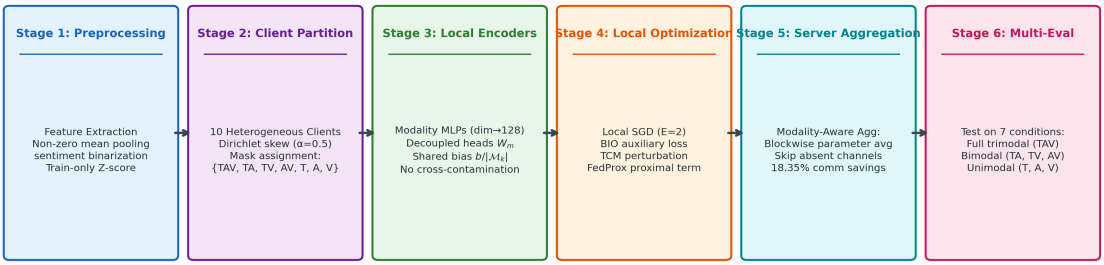


Figure 2: Implementation Pipeline Of The Complete Six Stages Of FIB-Fusion

The pipeline has six stages:

1. Data ingestion.
2. Mean pooling over non-zero time steps.
3. Standardization using training statistics only.
4. Dirichlet non-IID client partitioning.
5. Decoupled local optimization with information balancing.
6. Modality-aware server aggregation, followed by evaluation under all seven test conditions.

4.4 Modality-Aware Blockwise Server Aggregation

We split the parameters into modality-specific blocks:

$$\theta_m = \{\theta_m^{enc}, W_m\}\,, \tag{23}$$

with common bias b.
Let C_m^t be the set of clients involved in the training round with parameter modality m in round t.

$$C_m^t = \left\{k \in S^t \mid m \in M_k\right\}. \tag{24}$$

Each block is averaged over clients:

$$\theta_m^{(t+1)} = \sum_{k \in C_m^t} \frac{n_k}{\sum_{j \in C_m^t} n_j} \theta_{k,m}^{(t+1)} \tag{25}$$

for each modality $m \in \{T, A, V\}$.
If C_m^t is empty:

$$\theta_m^{(t+1)} = \theta_m^t. \tag{26}$$

The common bias b is aggregated over all participating clients.

5 Experimental Setup

5.1 Datasets and Preprocessing

We use two affective computing benchmarks.

5.1.1 CMU-MOSI

CMU-MOSI consists of 93 video monologues containing a total of 2199 utterances. We follow the standard protocol by removing the neutral utterances ($s = 0$):

- Training samples: 1,231
  - Positive: 679
  - Negative: 552
- Validation samples: 216
  - Positive: 124
  - Negative: 92
- Test samples: 656
  - Positive: 277
  - Negative: 379

Feature dimensions:

- Text: $D_T = 300$, from GloVe embeddings.
- Audio: $D_A = 5$, from COVAREP acoustic features.
- Visual: $D_V = 20$, from Facet visual action units.

5.1.2 CMU-MOSEI

CMU-MOSEI consists of 23,453 video utterances recorded by over 1000 speakers. After removing the neutral utterances we have:

- Training samples: 12,735
  - Positive: 8,014
  - Negative: 4,721
- Validation samples: 1,436
  - Positive: 931
  - Negative: 505
- Test samples: 3,620
  - Positive: 2,275
  - Negative: 1,345

Feature dimensions:

- Text: $D_T = 300$.
- Audio: $D_A = 74$.
- Visual: $D_V = 35$.

The feature vectors are obtained by mean pooling over non-zero time-steps giving a single vector of fixed length. The standardization statistics are only calculated on the training set and applied to the validation and test sets.

5.2 Federated Simulation Configuration

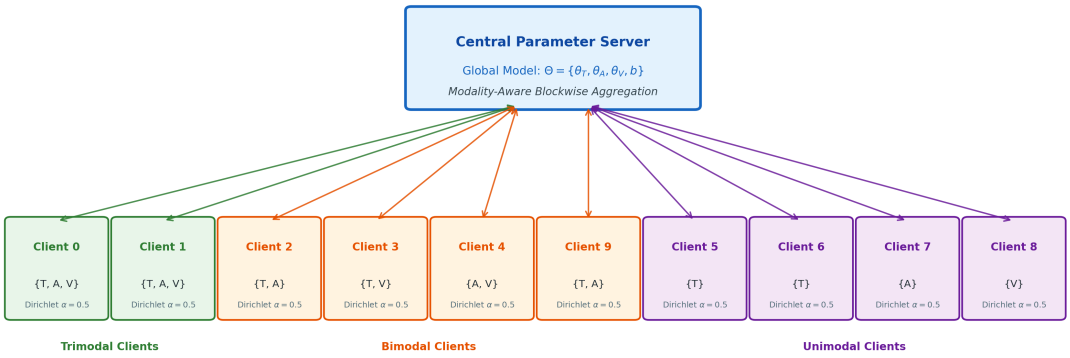


Figure 3: Architecture And Client Topology Of The Federated Learning

Ten edge clients are used, with:

- Trimodal: C0, C1.
- Bimodal: C2, C3, C4, C9.
- Unimodal: C5, C6, C7, C8.

Each client having a highly imbalanced class distribution due to Dirichlet label distribution with $\alpha = 0.5$. This simulates edge computing scenarios.

We have $K = 10$ clients in the network with:

- Clients 0 and 1 have all three modalities $\{T, A, V\}$.
- Clients 2 and 9 have $\{T, A\}$.
- Client 3 has $\{T, V\}$.
- Client 4 has $\{A, V\}$.
- Clients 5 and 6 have $\{T\}$ only.
- Client 7 has $\{A\}$ only.
- Client 8 has $\{V\}$ only.

Variation in statistics due to Dirichlet label distribution with concentration parameter $\alpha = 0.5$.

5.3 Training Hyperparameters

Each model trains for:

- 30 communication rounds.
- Full client participation, $C = 1.0$.
- $E = 2$ local epochs per round.

SGD with momentum 0.9, weight decay 10^{-4} and gradient clipping 5.0.

Learning rates were chosen from a grid search of the validation set.

- CMU-MOSEI: $\eta = 0.1$.
- CMU-MOSI: $\eta = 0.01$.

The proximal regularization strengths are:

- CMU-MOSEI: $\mu = 0.01$.
- CMU-MOSI: $\mu = 0.001$.

We repeat every experiment with three random seeds: 42, 43, and 44.

6 Results and Analysis

Table 1 and Figure 4 show the full results on CMU-MOSEI. Table 2 and Figure 5 show the results on CMU-MOSI.

6.1 CMU-MOSEI Findings

Table 1: CMU-MOSEI Main Experimental Results on Held-Out Test Set. Mean ± Standard Deviation over 3 Seeds.

Method	Acc (TAV)	Macro-F1 (TAV)	Bal-Acc (TAV)	Mean Miss-F1	Δ_{miss}	Disc. Ratio	Comm (MB)	Rounds to Thr
Centralized (CE)	0.7541 ± 0.0062	0.7276 ± 0.0046	0.7216 ± 0.0033	0.6325 ± 0.0237	0.0950 ± 0.0232	1.2804 ± 0.0047	24.73	14.67 ± 1.53
Centralized (IBML)	0.6482 ± 0.0287	0.5886 ± 0.0165	0.5913 ± 0.0136	0.4968 ± 0.0259	0.0918 ± 0.0094	1.0370 ± 0.0306	24.73	–
A0: Local Only	0.5949 ± 0.0549	0.5206 ± 0.0668	0.5917 ± 0.0302	–	–	–	0.00	–
A1: FedAvg	0.7679 ± 0.0293	0.7297 ± 0.0424	0.7202 ± 0.0404	0.6593 ± 0.0214	0.0704 ± 0.0382	1.3033 ± 0.0276	247.30	5.00 ± 1.73
A1b: FedProx	0.7713 ± 0.0296	0.7314 ± 0.0520	0.7232 ± 0.0494	0.6564 ± 0.0263	0.0750 ± 0.0397	1.2992 ± 0.0389	247.30	3.67 ± 2.08
A2: ModAgg	0.7588 ± 0.0342	0.7179 ± 0.0665	0.7145 ± 0.0632	0.6547 ± 0.0321	0.0632 ± 0.0384	1.3207 ± 0.0249	201.91	4.00 ± 0.00
A3: ModAgg + BIO	0.7418 ± 0.0294	0.7021 ± 0.0343	0.6939 ± 0.0320	0.6129 ± 0.0470	0.0892 ± 0.0147	1.3032 ± 0.0614	201.91	11.00 ± 1.41
A3b: ModAgg + TCM	0.7635 ± 0.0293	0.7233 ± 0.0593	0.7190 ± 0.0587	0.6506 ± 0.0255	0.0726 ± 0.0375	1.3369 ± 0.0515	201.91	7.33 ± 7.57
A4: ModAgg + IBML	0.7402 ± 0.0210	0.6938 ± 0.0287	0.6857 ± 0.0262	0.6169 ± 0.0254	0.0769 ± 0.0037	1.3284 ± 0.0484	201.91	21.50 ± 7.78
A5: IBML (No Warmup)	0.7570 ± 0.0351	0.7130 ± 0.0675	0.7096 ± 0.0652	0.6401 ± 0.0568	0.0729 ± 0.0124	1.2305 ± 0.1333	201.91	2.50 ± 0.71
A6: BIO + CE	0.7539 ± 0.0194	0.7188 ± 0.0264	0.7108 ± 0.0265	0.6238 ± 0.0176	0.0950 ± 0.0114	1.3038 ± 0.0699	201.91	13.50 ± 7.78
A1c: ModAgg + FedProx	0.7697 ± 0.0296	0.7333 ± 0.0529	0.7268 ± 0.0519	0.6591 ± 0.0282	0.0742 ± 0.0330	1.3126 ± 0.0182	201.91	3.33 ± 0.58
A7: FedProx + IBML	0.7741 ± 0.0159	0.7439 ± 0.0198	0.7345 ± 0.0194	0.6645 ± 0.0156	0.0794 ± 0.0258	1.2778 ± 0.0592	201.91	6.00 ± 1.00

On CMU-MOSEI, Method A7 (FedProx+IBML) performs best. It has a test accuracy of 0.7741 ± 0.0159, full modality macro-F1 score of 0.7439 ± 0.0198, and mean partial modality F1 score of 0.6645 ± 0.0156.

This outperforms the Centralized CE baseline (0.7276 ± 0.0046) by 1.63 percentage points in terms of full modality macro-F1 score. It also outperforms the Centralized IBML model (0.5886 ± 0.0165), by 15.53 points. There is quite a large standard deviation for the seed results on the federated methods, therefore whilst the gap between A7 and Centralized CE is significant there is some caution needed..

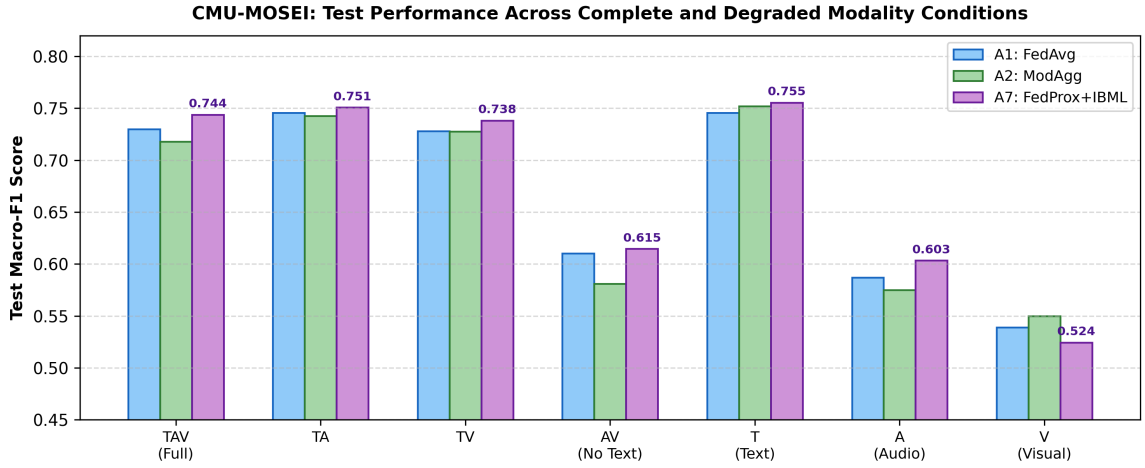


Figure 4: Performance of CMU-MOSEI on 7 Evaluation Settings (TAV, TA, TV, AV, T, A, V)

This figure shows test macro-F1 for each condition. Conditions that include text (TAV, TA, TV, T) hold up well. Removing text (AV) causes a sharp drop for every federated baseline. Among the methods shown, FedProx+IBML (A7) score highest in six of the seven conditions.

Centralized IBML performs poorly on CMU-MOSEI, mainly because recall on negative class drops to 0.3700 ± 0.0847. Adding the FedProx proximal term ($\mu = 0.01$ in A7) appears to steady local training under non-IID skew, and it lets model use the balancing objectives with no diverging.

Method A2 (modality-aware aggregation alone) has the smallest drop under missing modalities:

$$\Delta_{miss} = 0.0632 \pm 0.0384. \tag{27}$$

That compares 0.0950 for Centralized CE and 0.0918 for Centralized IBML. A2 also reaches the accuracy threshold quickly, in 4.00 rounds.

6.2 CMU-MOSI Findings

Table 2: CMU-MOSI Main Experimental Results on Held-Out Test Set. Mean ± Standard Deviation over 3 Seeds.

Method	Acc (TAV)	Macro-F1 (TAV)	Bal-Acc (TAV)	Mean Miss-F1	Δ_{miss}	Disc. Ratio	Comm (MB)	Rounds to Thr
Centralized (CE)	0.7870 ± 0.0074	0.7675 ± 0.0108	0.7632 ± 0.0129	0.6762 ± 0.0021	0.0913 ± 0.0105	1.3382 ± 0.0285	24.73	1.00 ± 0.00
Centralized (IBML)	0.7610 ± 0.0037	0.7362 ± 0.0054	0.7305 ± 0.0061	0.6658 ± 0.0037	0.0704 ± 0.0091	1.1739 ± 0.0258	24.73	1.33 ± 0.58
A0: Local Only	0.5857 ± 0.0554	0.5005 ± 0.0740	0.5879 ± 0.0351	–	–	–	0.00	–
A1: FedAvg	0.7647 ± 0.0338	0.7203 ± 0.0551	0.7117 ± 0.0505	0.5947 ± 0.0911	0.1256 ± 0.0379	1.3209 ± 0.1122	247.30	3.00 ± 0.00
A1b: FedProx	0.7640 ± 0.0327	0.7197 ± 0.0534	0.7111 ± 0.0491	0.5944 ± 0.0916	0.1253 ± 0.0402	1.3210 ± 0.1112	247.30	3.50 ± 0.71
A2: ModAgg	0.7736 ± 0.0173	0.7406 ± 0.0261	0.7309 ± 0.0263	0.6066 ± 0.0925	0.1340 ± 0.0666	1.3151 ± 0.0702	201.91	2.00 ± 0.00
A3: ModAgg + BIO	0.7746 ± 0.0165	0.7460 ± 0.0252	0.7380 ± 0.0269	0.6494 ± 0.0167	0.0966 ± 0.0149	1.2764 ± 0.0601	201.91	2.00 ± 0.00
A3b: ModAgg + TCM	0.7715 ± 0.0153	0.7399 ± 0.0281	0.7327 ± 0.0309	0.5988 ± 0.1022	0.1411 ± 0.0749	1.3397 ± 0.1181	201.91	2.00 ± 0.00
A4: ModAgg + IBML	0.7668 ± 0.0298	0.7319 ± 0.0522	0.7272 ± 0.0534	0.6533 ± 0.0165	0.0786 ± 0.0481	1.2546 ± 0.0658	201.91	2.00 ± 0.00
A5: IBML (No Warmup)	0.7610 ± 0.0336	0.7064 ± 0.0613	0.6992 ± 0.0523	0.6458 ± 0.0199	0.0606 ± 0.0517	1.2882 ± 0.0555	201.91	7.00 ± 3.61
A6: BIO + CE	0.7801 ± 0.0134	0.7555 ± 0.0160	0.7481 ± 0.0163	0.6503 ± 0.0263	0.1052 ± 0.0155	1.2748 ± 0.0480	201.91	2.00 ± 0.00
A1c: ModAgg + FedProx	0.7740 ± 0.0182	0.7400 ± 0.0266	0.7299 ± 0.0265	0.6062 ± 0.0931	0.1338 ± 0.0667	1.3125 ± 0.0739	201.91	2.00 ± 0.00
A7: FedProx + IBML	0.7680 ± 0.0221	0.7342 ± 0.0383	0.7275 ± 0.0414	0.6557 ± 0.0171	0.0785 ± 0.0334	1.2555 ± 0.0594	201.91	2.00 ± 0.00

On CMU-MOSI, Method A6 (BIO + CE) is the best federated method, with a test accuracy of 0.7801 ± 0.0134 and a macro-F1 of 0.7555 ± 0.0160. It sits just below the Centralized CE baseline (macro-F1 0.7675 ± 0.0108).

The modality-aware variants score higher than standard FedAvg and FedProx, though the gaps are smaller next to the variation between seeds:

- A2: ModAgg = 0.7406.

- A3: ModAgg + BIO = 0.7460.
- A6: BIO + CE = 0.7555.
- A7: FedProx + IBML = 0.7342.
- FedAvg = 0.7203 ± 0.0551.
- FedProx = 0.7197 ± 0.0534.

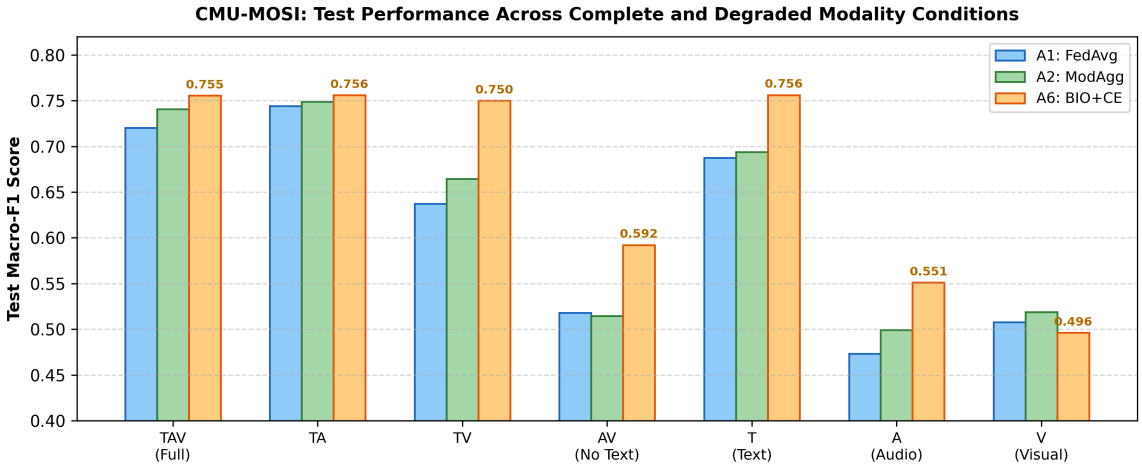


Figure 5: Performance of CMU-MOSI on 7 Evaluation Settings (TAV, TA, TV, AV, T, A, V)

This figure shows the gains from combining modality-aware aggregation with local BIO balancing (A6), especially in the audio-visual (AV) and audio-only (A) conditions.

Under missing modalities, standard FedAvg falls to a mean missing-modality F1 of 0.5947 ± 0.0911. Three methods do better:

- Method A3 (ModAgg + BIO) reaches 0.6494.
- Method A6 reaches 0.6503.
- Method A7 reaches 0.6557 ± 0.0171.

The best result is 6.1% higher than FedAvg.

In case of complete missing data (AV), using local BIO supervision (A6) improves the macro-F1 by a lot. It increases from 0.5181 (FedAvg) to 0.5923. This can be explained by the fact that the balancing gradients of the auxiliary task prevent the language modality from dominating the other modalities, however, this was not tested.

6.3 Comparing the Modalities

Table 3: Condition-Wise Test Macro-F1 Under Progressive Modality Degradation.

Condition	MOSEI: FedAvg (A1)	MOSEI: ModAgg (A2)	MOSEI: FedProx+IBML (A7)	MOSI: FedAvg (A1)	MOSI: ModAgg (A2)	MOSI: BIO+CE (A6)
TAV (Complete)	0.7297 ± 0.0424	0.7179 ± 0.0665	0.7439 ± 0.0198	0.7203 ± 0.0551	0.7406 ± 0.0261	0.7555 ± 0.0160
TA	0.7457 ± 0.0204	0.7426 ± 0.0358	0.7507 ± 0.0131	0.7440 ± 0.0343	0.7489 ± 0.0140	0.7560 ± 0.0146
TV	0.7279 ± 0.0376	0.7277 ± 0.0374	0.7382 ± 0.0196	0.6373 ± 0.1836	0.6644 ± 0.1496	0.7499 ± 0.0214
AV (No Text)	0.6104 ± 0.0090	0.5811 ± 0.0694	0.6147 ± 0.0144	0.5181 ± 0.0701	0.5146 ± 0.1178	0.5923 ± 0.0144
T (Text Only)	0.7457 ± 0.0269	0.7520 ± 0.0173	0.7555 ± 0.0115	0.6875 ± 0.1387	0.6939 ± 0.1112	0.7561 ± 0.0135
A (Audio Only)	0.5869 ± 0.0947	0.5749 ± 0.0418	0.6033 ± 0.0116	0.4732 ± 0.1856	0.4992 ± 0.1978	0.5511 ± 0.1266
V (Visual Only)	0.5390 ± 0.0539	0.5498 ± 0.0561	0.5245 ± 0.0801	0.5079 ± 0.1058	0.5187 ± 0.1157	0.4962 ± 0.0932

Table 3 shows how test macro-F1 changes across all seven conditions. Three things stand out.

1. **Text carries most of the signal, and removing it hurts the most.**

   The scores are high in general. On MOSEI they achieve around 0.73 and 0.76 in the conditions that include text. On MOSI they vary more by method, from about 0.64 to 0.76.

   Without text (AV), FedAvg score decreases to 0.6104 on MOSEI and 0.5181 on MOSI.

2. **Text is the strongest single channel, and audio usually beats visual.**

   The single modality models score anywhere between 0.69 and 0.76 depending on the model used. The audio only models score between 0.47 and 0.60 while the visual only models score between 0.50 and 0.55. While audio is outperforming visual in most cases, there are instances such as FedAvg on MOSI where audio is outperformed by visual (0.4732 vs. 0.5079).

   When using only visual features gives a result very close to chance. This is a given that interpreting sentiment from action units is difficult without accompanying words.

3. **Balancing helps the weaker channels.**

   On MOSI, adding local BIO balancing (A6) raises audio-only macro-F1 from 0.4732 (FedAvg) to 0.5511, a gain of 7.79 percentage points. This suggests that the auxiliary loss terms push the network to extract more from the acoustic features. The standard deviations on these audio-only numbers are large, though, so the result needs confirmation with more seeds.

6.4 Modality Discrepancy and Communication Cost

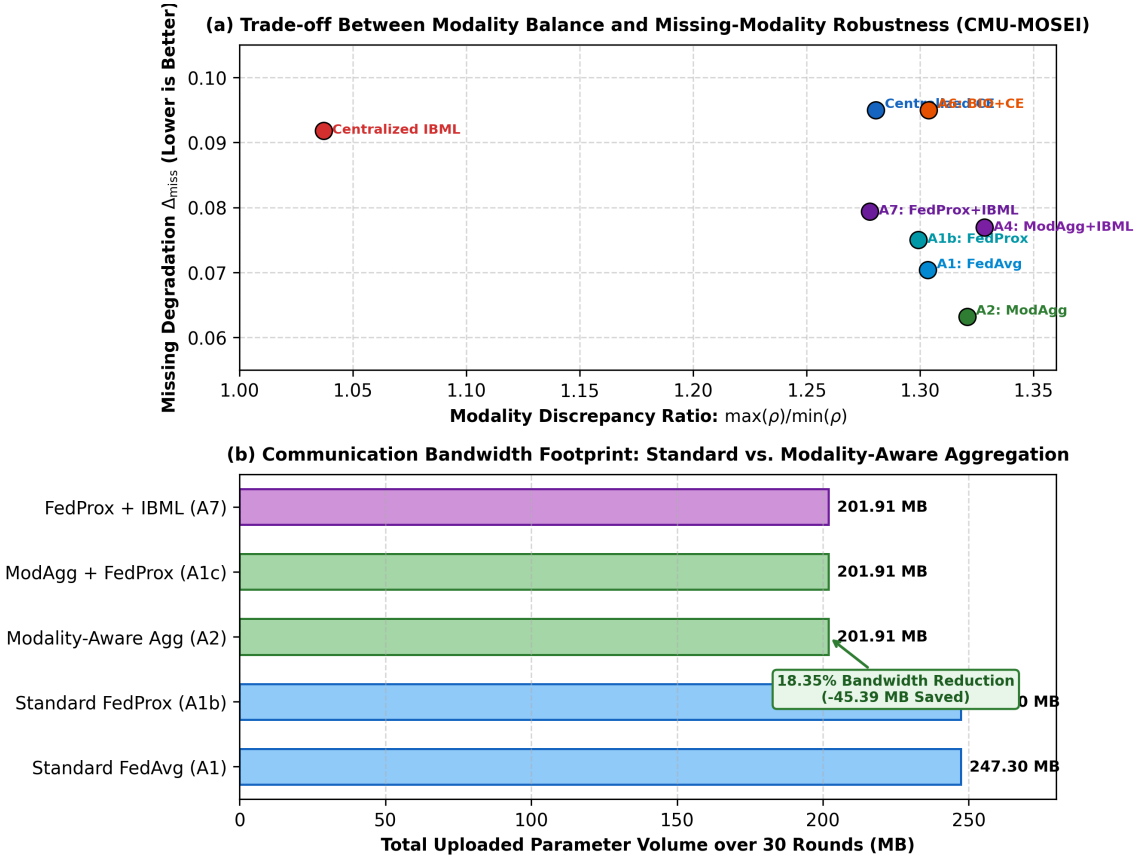


Figure 6: Modality Discrepancy Ratio and Communication Trade-off

The two subplots are stacked vertically so each stays readable.

(a) Modality discrepancy ratio ($\max(\rho)/\min(\rho)$) against the drop under missing modalities (Δ_{miss}) on CMU-MOSEI. Methods with more balanced modalities tend to hold up better when channels go missing.

(b) Total communication volume (MB) over 30 rounds. Modality-aware aggregation cuts it by 18.35%.

Figure 6 shows how modality discrepancy relates to communication cost.

6.4.1 Modality Discrepancy Ratio

The discrepancy ratio $\max(\rho)/\min(\rho)$ compares the confidence of the strongest and weakest modality for the true class.

On CMU-MOSEI, Centralized IBML reaches a ratio close to 1 (1.0370 ± 0.0306), so its modalities seem almost perfectly balanced, but its overall classification collapses. FedProx + IBML (A7) keeps the ratio of 1.2778 ± 0.0592 while keeping up on accuracy in the federated setting.

6.4.2 Communication Cost

The standard FedAvg and FedProx upload the complete set of model weights no matter the modalities of a client, which amounts to 247.30 MB over 30 rounds.

In modality-aware aggregation only parameters of the modalities a client actually has(θ_m for $m \in M_k$) are uploaded. The overall size is reduced to 201.91 MB, a reduction of 45.39 MB or 18.35%. Accuracy shows no clear loss from this: on MOSEI A2's macro-F1 is lower than FedAvg's, but the difference is well within seed-to-seed variation, and on MOSI A2 is higher.

7 Discussion

7.1 Decentralized IBML Versus Centralized Behavior

In centralized training, true-class confidence sums ρ_m are computed on minibatches sampled from a single fairly balanced pool.

In federated learning with non-IID partitions ($\alpha = 0.5$), labels are heavily skewed in each client. Let's take Client 4 with 100% positive data. Its confidence values ρ_m mainly reflect the class prior instead of the quality of the representation for each modality. Unimodal clients have no other modality to compare against and so they resort to plain cross-entropy.

These two effects likely explain the worse performance of IBML without a proximal term (Method A4) on MOSEI (macro-F1 0.6938) compared to the other modality-aware variants.

7.2 The Role of Proximal Regularization

On MOSEI, with FedProx added to IBML (A7) the macro-F1 increases from 0.6938 (A4, no proximal term) to 0.7439. This allows the two techniques to complement each other.

Local BIO loss can cause client weights to drift from the global consensus. The FedProx penalty:

$$\frac{\mu}{2} \left\| \Theta - \Theta^t \right\|_2^2 \tag{28}$$

drifting limits, so each client remains within a trust region around the global model, while still enjoying the balancing signal. The effect did not carry-over to MOSI where A7 (0.7342) and A4 (0.7319) were almost indistinguishable.

8 Limitations

1. **Temporal pooling.**

We averaged features over non-zero time steps to obtain static utterance-level vectors. This removes fine-grained word-level temporal cross-attention.

2. **Simulated edge environment.**

   Client partitions and modality masks were generated from benchmark splits. They were not observed in physical edge devices.

3. **No formal privacy guarantee.**

   The framework keeps raw video and audio off the server, but the model updates its transmits have no formal differential privacy guarantee.

9 Conclusion and Future Work

This paper studied FIB-Fusion on CMU -MOSI and CMU -MOSEI when clients missing sensors and label distributions are skewed.

Modality aware block wise aggregation stops absent channels from diluting active ones and cuts communication by 18.35%.

On CMU MOSEI, combining FedProx and IBML A7 gave the best results :

- Full modality F1: 0.7439.
- Missing modality F1: 0.6645.

For this data set it scored above the standard baselines.

On the smaller CMU MOSI dataset, local BIO supervision with cross entropyA6 did the most for strengthening the non text encoders.

Future work will look at transformer-based temporal encoders and at adding differential privacy.

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Links:
GitHub: https://github.com/NeeharikaReddy7726/ML_Research_project
LaTeX: https://www.overleaf.com/project/6ae9c084f4ac5aa02e02ff27/share#be0db588832ddd42b7f44d555
Declarations * Funding: No funding was received to assist with the preparation of this manuscript. * Conflict of Interest / Competing Interests: The authors declare that they have no conflict of interest. * Ethics Approval and Consent to Participate: Not applicable. * Consent for Publication: Not applicable. * Data Availability: The datasets referenced in the related works are publicly available, and their URLs are provided in the reference section. * Materials Availability: Not applicable. * Code Availability: Not applicable. * Author Contributions: Author 1 contributed to the conceptualization, methodology, data analysis, and preparation of the original draft. Author 2 contributed to the critical review, editing, supervision, and overall refinement of the manuscript. Both authors read and approved the final manuscript.